THEME 1: FUNCTIONS

Unit 1.1 Numbers, intervals and inequalities

Source: Stewart, Appendix A pp A2 - A6.

Learning objectives

On completion of this unit you should

1. know the definitions of the following number systems: the set $\mathbb{N}$ of all natural numbers, the set $\mathbb{Z}$ of all integers, the set $\mathbb{Q}$ of all rational numbers, the set $\mathbb{I}$ of all irrational numbers and the set $\mathbb{R}$ of all real numbers;
2. be able to use set notation;
3. be able to use the word "and" for the intersection of two sets;
4. be able to use the word "or" for the union of two sets;
5. be able to find the union/intersection of two sets;
6. be able to use interval notation;
7. know the rules for inequalities and be able to use it;
8. be able to solve inequalities.

Remarks

1. The set $\mathbb{N}$ of all natural numbers is the set $\mathbb{N} = \{1, 2, 3, \dots\}$.
2. Pay special attention to Table [1] on p A4. You must be able to describe an interval in interval notation and in set notation.
3. **Use a graph to solve inequalities where possible**. It is quicker and easier than the algebraic approach.
4. You must know and be able to use the following formula:

$$ax^2 + bx + c = 0 \Longrightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Assigned Problems

Stewart, Exercises A p A9: Numbers 17, 19, 25, 27, 29, 35, 39, 42, 58, 60.

Unit 1.2 The absolute value

Source: Stewart, Appendix A pp A6 - A9.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the absolute value.
2. be able to write down and use the properties of the absolute value.
3. be able to solve inequalities with absolute values.
4. be able to sketch the graph of $y = |f(x)|$.

Remarks

1. If $x \in \mathbb{R}$ then $\sqrt{x^2} = |x|$.
2. When solving inequalities with absolute values, use the theorems in [6] on p A7.
3. Omit the triangle inequality on p A8 and Example 9 on p A9.
4. You can also use the following theorem: If x and y are real numbers, then $|x| \leq |y|$ if and only if $x^2 \leq y^2$.

Assigned Problems

1. Stewart, Exercises A p A9: Numbers 1, 3, 7, 9, 12, 45, 47, 49, 51, 53, 55, 56.
2. Sketch the graphs of:
   a. $y = |x - 3|$
   b. $y = |x^2 - 16|$
   c. $y = |3x - 2|$

Unit 1.3 What is a function?

Source: Stewart, Section 1.1 pp 8 - 21.

Learning objectives

On completion of this unit you should

1. be able to identify a function.
2. be familiar with the four possible ways of representing a function.
3. be able to find the domain of a function.
4. be able to find the range of some functions by using the graph of the function.

5. know how to test whether a curve in the xy-plane is the graph of a function by using the Vertical Line Test.
6. be familiar with piecewise-defined functions.
7. be able to sketch the function $y = |f(x)|$ if the graph of the function $y = f(x)$ is given.
8. be able to write down and use the definition of an even function.
9. be able to write down and use the definition of an odd function.
10. be able to write down and use the definition of an increasing function.
11. be able to write down and use the definition of a decreasing function.

Assigned Problems

Stewart, Exercises 1.1 p 17: Numbers 1, 3, 7, 9, 25, 31, 33, 35, 41, 42, 45, 47, 51, 55, 71.

Unit 1.4 Radian measure and trigonometric functions

Source: Stewart, Appendix D pp A24 - A35.

Learning objectives

On completion of this unit you should

1. be able to use radian measure.
2. be able to write down and use the definitions of the six trigonometric functions.
3. be able to sketch the graphs of the functions $y = \sin x,\ y = \cos x$ and $y = \tan x$.
4. be able to write down and use some trigonometric identities.
5. be able to solve trigonometric equalities and inequalities.

Remarks

1. Omit Example 5.
2. Memorize identities 7, 8, 13, 14, 16 and 17 on pp A28 - A29. The others can be easily deduced from these, make sure you can do this.
3. You must be able to calculate all the function values in the table on p A27.
4. You must be able to find the exact trigonometric function values for the angles $\pi, \frac{\pi}{2}, \frac{\pi}{3}, \frac{\pi}{4}, \frac{\pi}{6}$ and any related angles without using a calculator.
5. **We always use radian measure when we work with trigonometric functions.**

Assigned Problems

1. Stewart, Exercises D p A33: Numbers 3, 9, 19, 20, 25, 26, 31, 32, 34, 45, 47, 59, 69, 71, 72, 75, 76.

2. Find $\sin(75\pi)$ and $\cos\left(\frac{17\pi}{4}\right)$ without using a calculator.

Unit 1.5 A catalogue of essential functions

Source: Stewart, Section 1.2 pp 21 - 36.

Learning objectives

On completion of this unit you should

1. be able to identify the following functions: linear functions, power functions, polynomials, algebraic functions, rational functions, exponential functions and logarithmic functions.
2. be able to sketch the graphs of linear functions, power functions, some polynomials, exponential functions and logarithmic functions.

Remark: Omit Examples 2, 3 and 4.

Assigned Problems

Stewart, Exercises 1.2 p 33: Numbers 1, 3, 4, 6, 8, 11, 13, 14.

Unit 1.6 New functions from old functions

Source: Stewart, Section 1.3 pp 36 - 45.

Learning objectives

On completion of this unit you should

1. be able to transform a function through translations: vertical and horizontal shifts, stretching, compressing and reflecting.
2. be able to combine two functions f and g to form new functions $f+g$, $f-g$, fg and $\frac{f}{g}$ and be able to determine the domains of these functions.
3. be able to form the composite functions $f \circ g$ and $g \circ f$ for given functions f and g and be able to determine the domain of the new functions.
4. be able to decide what the functions f and g are if the composite function $f \circ g$ is given.

Remarks

1. You should be able to sketch the new transformed function by using the graph(s) of the old function(s).
2. Omit Example 4 on p 39.
3. The functions $y = \sin(Bx)$ and $y = \cos(Bx)$ have period $\dfrac{2\pi}{|B|}$.

Assigned Problems

1. Stewart, Exercises 1.3 p 42: Numbers 1, 3, 6, 7, 13, 14, 33, 36, 37, 43, 45, 47, 48, 51, 57, 65.

2. Stewart, Exercises D p A35: Numbers 83 and 87.

3. Sketch the following functions on the interval $[-2\pi, 2\pi]$:

   a. $f(x) = 2\sin x$

   b. $g(x) = \sin(2x)$

   c. $h(x) = 2 - \sin x$

Unit 1.7 Exponential functions

Source: Stewart, Section 1.4 pp 45 - 53.

Learning objectives

On completion of this unit you should

1. be able to write down and use the equation of an exponential function.
2. know that e is the symbol for a significant irrational number.
3. be able to sketch the graphs of the following exponential functions: $y = a^x, a > 1$ and $y = a^x, 0 < a < 1$ and $y = e^x$ and $y = e^{-x}$.
4. be able to sketch the graphs of shifted and translated exponential functions.
5. be able to write down and use the laws of exponents.

Remarks

1. You do not have to know how a^x, with x an irrational number, is defined (if interested, see the explanation on pp 50 - 51).
2. Omit Example 6 on p 51.

Assigned Problems

Stewart, Exercises 1.4 p 52: Numbers 1, 3, 13, 15, 16, 19, 21, 29(a) to (c).

Unit 1.8 Inverse functions

Source: Stewart, Section 1.5 pp 54 - 57.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a one-to-one function.
2. know how to test whether a function is one-to-one by using the Horizontal Line Test.
3. be able to write down and use the definition of an inverse function f^{-1}.
4. be able to write down the relationship between the domains and ranges of f and f^{-1}.
5. be able to write down and use the cancellation equations of the composites of f and f^{-1}.
6. be able to find the inverse function of a one-to-one function f.
7. be able to explain the relationship between the graphs of f and f^{-1}.

Remarks

1. The definition of a one-to-one function can be written in two ways:
   A function $f : A \to B$ is one-to-one if, for all $x_1, x_2 \in A$, $x_1 \neq x_2 \Longrightarrow f(x_1) \neq f(x_2)$.
   We can also say: A function $f : A \to B$ is one-to-one if $f(x_1) = f(x_2) \Longrightarrow x_1 = x_2$.
2. The three steps to find a inverse function is given on p 56.

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 5, 7, 11, 17, 20, 25.

Unit 1.9 Logarithmic functions

Source: Stewart, Section 1.5 pp 57 - 61.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the logarithmic function, $y = \log_a x$.
2. be able to write down and use the laws of logarithms.
3. be able to write down and use the definition of the natural logarithmic function, $y = \ln x$.
4. be able to write down and use the properties of the natural logarithm.
5. be able to use the cancellation equations of the composites of $y = \ln x$ and $y = e^x$ to solve equations and inequalities.
6. be able to sketch the graph of the function $y = \ln x$.

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 27, 28, 30, 40, 42, 57, 59, 62, 63, 64.

Unit 1.10 Inverse trigonometric functions

Source: Stewart, Section 1.5 pp 61 - 64.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the inverse trigonometric functions $y = \arcsin x,\ y = \arccos x$ and $y = \arctan x$.
2. be able to write down and use the properties of the inverse trigonometric functions $y = \arcsin x,\ y = \arccos x$ and $y = \arctan x$.
3. be able to sketch the graphs of the functions $y = \arcsin x, y = \arccos x, y = \arctan x$.
4. be able to use the cancellation equations of the composites of $f$and f^{-1} to solve equations.

Remark

Omit the definitions of the other three inverse trigonometric function ([12] on p 64).

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 69, 70, 71, 72, 74, 75, 76, 77, 78, 81.
